

COMP 325 CAT II

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Do in groups of maximum three

1. **Write a Prolog program to input a number and then output either its square or its cube depending on the user's decision. This processing should be repeated as long as the input number is positive.**

start :-

```
write('Enter a number: '), read(N),  
(N > 0 ->  
    (write('Enter "s" for Square or "c" for Cube: '), read(Choice),  
      calculate(N, Choice),  
      start) ;  
    write('Number is not positive. Goodbye!'))  
).
```

```
calculate(N, s) :- Res is N * N, write('Result (Square): '), write(Res), nl.  
calculate(N, c) :- Res is N * N * N, write('Result (Cube): '), write(Res), nl.  
calculate(_, _) :- write('Invalid choice!'), nl.
```

2. **Write down the output of the following Prolog program (go). Thus explain the work of go.**

go:-compute(10,3).

compute(A,B):- (A>B;show(A,B)), (A<B;T is B+3), (A<B;compute(A,T)).

show(A,B):-T is B-3,K is A-T,write(K).

Output: 1

true

Explanation :

The program finds the remainder of \$A\$ divided by \$3\$, but it does so through recursion.

1. compute(10,3) starts. Since \$10 > 3\$ is true, it skips the show predicate initially.
2. It increments \$B\$ by 3 (\$3\$ to \$6\$ to \$9\$ to \$12\$).
3. When \$B\$ becomes 12, the condition \$A > B\$ (\$10 > 12\$) becomes False.

4. This triggers show(10, 12).

5. Inside show, it takes the previous value of \$B\$ (which was \$12 - 3 = 9\$) and subtracts it from \$A\$ (\$10 - 9 = 1\$).

6. It then writes the result: 1.

3. Fill in the test table for the Prolog program (get) shown below. Thus, explain the work of get.

The Program	Test table		
	Inputs		Outputs
	A	B	
	6	3	End
	3	6	3456
loop(M, K).	6	6	6

Explanation: The program uses read to take two numbers. If \$A > B\$, the first part of the disjunction (\$A < B\$; write('End')) fails the check and executes the write command. If \$A \leq B\$, the second part (\$A > B\$; loop(A, B)) fails the check and triggers the recursive loop to print the range.

4. **Functions receiving values:**

E.g: compute() below receives one parameter A. And compute_1() receives two parameters.

compute(A):- (A<0; compute_1(A, 0)), (A>=0; compute_1(0, -1)).

compute_1(0, M):- write(M).

compute_1(B, T):- C is (B+T), D is (B-1), compute_1(D,C).

Required: Write the outputs of the following queries.

(i) **compute(-4).**

Output: -1

Reason: Since \$-4 < 0\$ is true, it skips the first compute_1. However, the second part (\$A \geq 0\$; compute_1(0, -1)) fails the first check, calling compute_1(0, -1), which simply writes -1.

(ii) **compute(3).**

Output: 6

Reason: It calculates \$3 + 2 + 1 + 0 = 6\$.

(iii) **compute(0).**

I.e. calling compute and passing a value.

Output: 0

Reason: It calls compute_1(0, 0), which writes 0.

5. Functions receiving, returning values:

(i) Define a Prolog function equal(A, B, T) that receives two numbers (A and B), and returns (as the third parameter) the value of T as either 1 (if A equals B), or 0 (if otherwise).

Answer: Using prototypes of equal()

equal(A, B, T):- A==B, T is 1.

equal(A, B, T):- A\==B, T is 0.

Alternative: Using the operator :

equal(A, B, T):- (A\==B; T is 1), (A==B; T is 0).

(ii) Write the result (returned) of the following queries based on the above function.

(I) equal(5, 8, P). P = 0. (5 \== 8, so second clause runs. T is 0 binds P to 0.)

(II) equal(8, 8, A). A = 1. (8 == 8, so first clause runs. T is 1 binds A to 1.)

(III) equal(2, 2, t). false (The expected return is the number \$1\$; the atom t does not match).

(IV) equal(2, 5, 0). true (The function determines \$T=0\$, which matches the provided \$0\$).

6. Functions receiving, returning values:

The following Prolog program (i.e. func() with 3 prototypes) receives two numbers and returns (as the third parameter) their difference i.e. either A-B (if A>B) or B-A (if B>A) or 0 (if A=B).

func(A, B, C):-A>B, C is A-B.

func(A, B, C):-B>A, C is B-A.

func(A, A, C):-C is 0.

Exercise

(i) Rewrite the program using the operator ;

func(A, B, C) :-

(A > B, C is A - B) ;

(B > A, C is B - A) ;

(A == B, C is 0).

(ii) Write the outputs of the following queries.

(I) ?- func(12, 4, B). B = 8. (12 > 4 → C is 12-4 → B=8.)

(II) ?- func(5, 5, T). T = 0. (5>5 false, 5>5 false → falls to else → T is 0.)

(III) ?- func(3, 6, 0). false (Calculated difference is 3, not 0).

(IV) ?- func(25, 35, A). A = 10. (35 > 25 → A is 35-25 → A=10.)

(V) ?- func(3, 6, t). false (Calculated difference is 3, which does not unify with the atom t).